Katherine L. Titterton

Scientist-engineer and technical storyteller in neuroscience, imaging and AI

Analytical researcher in high-content imaging. Develops and scales assays until the signal is clean, then extracts what the images hold: from *in vitro* neurodegeneration models to AI-ready datasets. Learned to code at night before AI could; now builds with people and AI agents, for people.

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EXPERIENCE

Axiom Bio Founding Member of the Technical Staff — San Francisco, CA — 2023–2026

- Oversaw generation and QC of 10M+ Cell Painting images of 100,000+ compounds (nearly every commercial bioactive), the training data for models that predict bioactivity profiles from structure alone, before synthesis.
- Invented two low-noise algorithms for morphology change, ~10× more sensitive than conventional cytotoxicity assays, which made them top features of a 0.92 AUC liver-injury (DILI) model; they also caught batch drift.
- Led image labeling specs, QC and PyTorch U-Net segmentation of the dataset, benchmarked against Segment Anything (SAM); models in production for 1+ year.

Nat Commun 2026 [4] Cell Syst 2026 [3] bioRxiv 2026 [P1]

Insitro Associate Scientist III, Arrayed Screening — South San Francisco, CA — 2022–2023

- Adapted and ran high-content phenotypic screens for tuberous sclerosis complex and ALS on automated platforms (BioTek, HighRes Biosolutions), from assay development to 384-well feeding, staining and imaging.
- Automated QC and analysis of hundreds of 384-well plates with self-taught Python and R.
- Built a digital phase-contrast imaging modality and segmentation for label-free tracking of neuron growth.
- Trained two screen operators.

MIT Technical Associate II, Synthetic Neurobiology Group — Cambridge, MA — 2021–2022

- Optimized multiplexed expansion revealing (mExR) for 7+ immunostaining rounds: 30+ proteins at nanoscale in 16× expanded brain tissue.
- Ran the Boyden Lab's shared ImageJ/Fiji and MATLAB image-analysis pipelines and helped optimize their performance.

Nat Commun 2024 [6]

AbbVie Associate Scientist II, Foundational Neuroscience Center — Cambridge, MA — 2018–2021

- Cultured primary neurons weekly for 3+ years, supporting 10+ tau team scientists; scaled the culture from manual to automated by onboarding Integra liquid handlers.
- Developed high-content tau aggregation assays in primary murine and human iPSC neurons (Revvity Opera, Harmony), with R dose-response pipelines that ranked tau immunotherapy candidates.
- Screened anti-tau antibodies by high-content analysis in primary neurons (SBI2 2019).
- Tested next-gen microfluidic devices, keeping primary murine neurons alive 3+ months to model tau spread (SfN 2021).

Mol Neurodegener Adv 2026 [1] iScience 2026 [2] J Biol Chem 2025 [5] iScience 2022 [7]

Massachusetts General Hospital Research Technician I — Boston, MA — 2017–2018

- Recorded voltage-clamp currents from GABA-A receptors in *Xenopus* oocytes to study how anesthetics act.

Eur J Med Chem 2019 [8]

EDUCATION

Colgate University B.A. Cellular Neuroscience, *cum laude* Hamilton, NY 2013–2017
Minor in Mathematical Systems Biology. Co-captain, Division I Track & Field.

Harvard Extension School Coursework Cambridge, MA 2017–2020
Undergraduate and graduate: Python, computer vision, social medicine, biochemistry.

PUBLICATIONS

[1] Yanamandra K, Romanul N, Patel H, Sanders H, Chang R, Darcy A, Brown N, Kane C, Wu HY, Jagtap K, Chancellor S, Morozova O, Xi M, Hogan C, **Titterton K**, Wu JW, Manos J, Karran E, Dellovade T, Langlois X. Conformation-specific anti-pathological tau antibody raised against AD brain-derived seed-competent tau reduces tau pathology in transgenic mice. *Mol Neurodegener Adv*. 2026;2(1):43. doi:10.1186/s44477-026-00049-6 ABBVIE

[2] Sebastian RM, Kupp R, Nanavati D, Parfentev I, Krzystek TJ, Romanul N, Yanamandra K, Preiss CN, Manos JD, Wu HY, Volkova A, Crescenti S, **Titterton K**, Welker AM, Townsend M, Karran E, Langlois X, Wu JW. SynCAR platform for capturing neuronal synaptic proteopathic seeds. *iScience*. 2026;29(7):116362. doi:10.1016/j.isci.2026.116362 PMID 42325569, PMC13276770 ABBVIE

[3] Ewald JD, **Titterton KL**, Bäuerle A, Beatson A, Boiko DA, Cabrera ÁA, Cheah J, Cimini BA, Gorissen BL, Harrill J, Jones TR, Karczewski KJ, Crute CE, Rouquie D, Seal S, Weisbart E, White B, Carpenter AE, Singh S. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Syst*. 2026;17(5):101566. doi:10.1016/j.cels.2026.101566 PMID 41903531 AXIOM BIO

[4] Seal S, Dee W, Shah A, Cerisier N, Zhang A, Miglietta E, **Titterton K**, Cabrera ÁA, Boiko D, Beatson A, Slabaugh G, Taboureau O, Carreras Puigvert J, Singh S, Spjuth O, Bender A, Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*. 2026;17(1). doi:10.1038/s41467-026-68725-5 PMID 41651839, PMC12988037 AXIOM BIO

[5] Wu JW, **Titterton K**, Sebastian R, Romanul N, Yanamandra K, Mastis B, Chang R, Khasnavis S, Kwon T, Chai Y, Manos JD, Preiss CN, Hu M, Welker AM, Liao F, Dellovade T, Karran E, Langlois X. Development of a robust, physiologically relevant neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem*. 2025;301(12):110853. doi:10.1016/j.jbc.2025.110853 PMID 41135680, PMC12670552 ABBVIE

[6] Kang J, Schroeder ME, Lee Y, Kapoor C, Yu E, Tarr TB, **Titterton K**, Zeng M, Park D, Niederst E, Wei D, Feng G, Boyden ES. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nat Commun*. 2024;15(1):9722. doi:10.1038/s41467-024-53729-w PMID 39521775, PMC11550395 MIT

[7] Manos JD, Preiss CN, Venkat N, Tamm J, Reinhardt P, Kwon T, Wu J, Winter AD, Jahn TR, Yanamandra K, **Titterton K**, Karran E, Langlois X. Uncovering specificity of endogenous TAU aggregation in a human iPSC-neuron TAU seeding model. *iScience*. 2022;25(1):103658. doi:10.1016/j.isci.2021.103658 PMID 35072001, PMC8761709 ABBVIE

[8] Wu B, Jayakar SS, Zhou X, **Titterton K**, Chiara DC, Szabo AL, Savechenkov PY, Kent DE, Cohen JB, Forman SA, Miller KW, Bruzik KS. Inhibitable photolabeling by neurosteroid diazirine analog in the β3-subunit of human heteropentameric type A GABA receptors. *Eur J Med Chem*. 2019;162:810-824. doi:10.1016/j.ejmech.2018.11.019 PMID 30544077, PMC6912599 MGH

PREPRINT

[P1] Ollitrault G, Mitchell CA, Ferguson SS, Roe AL, White B, **Titterton K**, Ewald J. Cell Painting and machine learning for mechanistic profiling of hepatotoxic botanicals and their constituents. *bioRxiv*. 2026. doi:10.64898/2026.10.05.756246 AXIOM BIO

TALKS & POSTERS

Titterton K. Using AI to Combine Exposure, Metabolism, and Liver Function to Model DILI in Modern Molecules. Talk, *American College of Toxicology (ACT) Annual Meeting*, Phoenix, AZ, 2025. replay

Titterton K, Boiko D, Cabrera ÁA, Bäuerle A, Tyagi A, Saneinejad S, Beatson A, White B. Accurate and interpretable *in silico* clinical risk assessment for drug-induced liver injury (DILI) from molecular structure. Poster, *Society of Toxicology (SOT) Annual Meeting*, 2025.

Titterton K, et al. Development of a tau propagation neuronal microfluidic cell culture model for assessing inhibitors of tau propagation. Poster, *Society for Neuroscience (SfN) Global Connectome*, 2021.

Titterton K, et al. High content analysis of tau aggregation in primary neurons: screening anti-tau antibodies and modeling tauopathy progression *in vitro*. Poster, *SBI2 High Content Meeting*, Harvard Medical School, 2019.

Investigating RNAi as a potential therapeutic for Huntington's disease in a transgenic mHTT145Q zebra finch (*Taeniopygia guttata*) model. Poster, *Colgate University Neuroscience Thesis Presentations*, 2017.

Atopy and helminth infection in Ethiopian schoolchildren. Poster, *Colgate Summer Research Symposium*, 2016.

PDE2 regulation of β-arrestin signaling. Poster, *Systems Biology Center New York Summer Trainee Symposium*, Icahn School of Medicine at Mount Sinai, 2015.

PROJECTS

Datum Line AI × bio startup: one map of every lab's controls, to tell if cells look normal `code` 2026
On public control wells from many labs, a classifier names the lab from controls alone, and a lab's drift predicts disagreement on known answers.

Every line is a loop A periodic table for biology `code` 2026
Interactive essay that reads the published BioModels and counts their circuits into a table: every switch has a self-feeding loop and every clock a brake. Live figures of the lac switch, a circadian clock and chemotaxis.

Overdose Watch Public tracker kept current by a weekly agent `code` 2026
Official overdose death counts (CDC, state and county health departments, the SF medical examiner) and primary-sourced incident reports on one map and timeline, refreshed every Monday by a scheduled Claude Code agent.

Vanitas / Veritas AI slop as vanitas `code` 2026
A 4K video loop melting vanitas portraits with open-source diffusion (Stable Diffusion 1.5, LCM-LoRA, Real-ESRGAN) on a laptop, with an interactive web piece.

treble DJ-set software: a playlist in, a mixed set out `code on request` 2026
Beat-aligned transitions, a listening studio and a taste loop, where every kept transition teaches the engine. Hear its sets p[i]lot (picked by hand), p[ai]lot (full algorithm) and b[u][r]ning.

claude-code-learns Graph-engineered agent harness: corrections become skills `code` 2026
Each correction becomes a skill; automated checks confirm every skill parses and loads when it should.

Bare Metal Solo generative-art exhibit, Human Tech Week SF, with a post-hoc analysis 2025

Vivarium 2.0 Automated poison-dart-frog habitat with misting and live video 2024

OpenTronsformers LLM copilot for the OT-2 lab robot; 2nd-place team, BioHack NYC 2023

HONORS & AWARDS

2nd place, BioHack NYC OpenTronsformers, LLM-driven lab-robot automation 2023

Dorothy-Waltham Award for Athletic Leadership Colgate University 2017

SKILLS

Code & agents: Python (NumPy, SciPy, scikit-learn, pandas, matplotlib, seaborn, Jupyter); R (tidyverse, ggplot2, drc, platetools); Git and GitHub; Claude Code (skills, hooks, scheduled agents); agent harnesses and graph engineering; LLM evals; Cursor; HTML, CSS and SVG for interactive pages

ML & data: PyTorch; U-Net segmentation (Cellpose and SAM evaluated); labeling specs and dataset QC; feature engineering; image embeddings (visualization, clustering, normalization); dose-response curve fitting (Prism, R, Python); univariate and multivariate statistics; design of experiments

Imaging & signal: high-content and confocal microscopy (Opera Phenix, Operetta, Harmony); expansion microscopy; label-free phase-contrast timecourse imaging; image formation and instrument artifacts; ImageJ/Fiji macros; MATLAB; voltage-clamp electrophysiology

Lab & automation: primary murine and human iPSC neuron culture; microfluidic culture models; immunocytochemistry (ICC) and immunohistochemistry (IHC); assay development; high-throughput, high-content screening; stereotaxic surgery and tissue microdissection; liquid handlers and schedulers (Opentrons, Integra, BioTek, Labcyte Echo, HighRes Biosolutions, Green Button Go); Benchling

Story & design: data visualization and figure design; pitch-deck and web design; generative art (Midjourney, Stable Diffusion); scientific storytelling for talks and posters

MEMBERSHIPS

Society of Toxicology, Society for Neuroscience, Society of Biomolecular Imaging and Informatics